

REPORT 2.6 - The lie card fits the phone (build v26)

Opus, 2026-09-16, spawned by Fable for one job. Tree at e5f55fb, clean, at the spawn. His words on it:

"If the bottom row is cut off, v25 stays off the course and the numbers come back here." (the phone check, docs/handoff/NEXT_CHAT_2026-09-15.md)

and this morning, with the photograph of v25 on hole 10 with the lie card up:

"3. Build it."

1. The answer, in plain words

The bottom row was cut because his phone gives the app a 728 px page, not the 780 px screen v25 was measured against. v26 gives the lie grid 54 px back and it now clears the footer by 25 px on that 728 px page.

Nothing was taken out of the footer. MARK SHOT, ENTER PUTTS, ENTER YARDAGES, MARK CUP, PENALTY and UNDO are all still there, in the same order. They are shorter: MARK SHOT 88 -> 80 px and the three below it 60 -> 52 px, the gaps between them 14 -> 12 px, and the lie card's own padding 6 px tighter. MARK SHOT is still the biggest control on the screen by 28 px, and nothing is smaller than 48 px, which is the smallest target Android will call a target. The 16 px dead zone between PENALTY and UNDO - the pair a wet thumb must never confuse - is untouched.

One state is still short of the fold, Section 6: if a shot is marked while the hint reads *"On the green: MARK CUP when you walk behind the hole, then enter the putts."* that sentence wraps to three lines and takes 38 px off the body, leaving the bottom row 11.5 px short. It needs a shot already marked GREEN, no cup yet, and MARK SHOT pressed instead of MARK CUP. Under v25 the same state was 56 px short. The fix is to shorten that sentence and the "Cup marked" one, and those are his words, so they are his call.

2. Why the screen is not the page (n = 1 photograph, read in pixels)

The S26 is 1080 x 2340 at density 480 and font_scale 1.0, which is 360 x 780 CSS px - the size REPORT 2.4 passed v25 at, with 24 px to spare. But Android keeps the top and bottom of that screen for itself. Sampling the photograph down its left edge (PIL, the original 1080 x 2340 file):

device px	what is there	CSS px
0 - 110	status bar, painted with the theme-color #f2efe4	0 - 37
111	the hud's surface starts - the top of the page	page y = 0
291 - 293	the hud's bottom border	hud is 61 tall
504 - 506	the hole nav's bottom border	hole nav is 71
1004 - 1006	the footer's border-top	body bottom = 297.7
2295	the gesture bar - the bottom of the page	page = 728 tall

So the page is $(2295 - 111) / 3 = 728$ CSS px. The check that says the reading is right: the footer measures **430.3 px** on that photograph, against **430.3 px** measured here at 360 px wide under v25. Two independent numbers, one layout.

728 - 61 hud - 71 hole nav - 430 footer leaves a **167 px body** for a 140 px lie grid that starts 56 px down: **29 px short**, which is what the photograph shows - the SAND / RECOVERY / GREEN row cut at about its middle.

REPORT 2.4 called this, exactly: *"There are 24 px to spare at 360 x 780 on the PC; if your phone's page is more than 24 px shorter, the SAND / RECOVERY / GREEN row will be cut."* It is 52 px shorter.

The photograph is v25: the card is first in the body with no banner above it, and the lie label is one line at 13 px - both are v25's doing, and the footer height matches v25's to a third of a pixel.

3. The test that was green while his phone was cut

The fold test added in 2.2 and passed in 2.4 was passing on a layout no phone has. . footer is clamped to max-height: 78dvh, and dvh is 0 in a hidden browser pane - the same zero that made zoneOf misread the lock zones in 2.2. Clipped to zero, the footer collapsed to its 19 px of padding, the body took the other 400 px, and the test read a body bottom of **709 px at 360x728** where a phone reads 298.

test/run.js now restates that clamp from the harness's own page height, and the numbers it reads match the phone: body bottom **297.72** in the harness against **297.7** off the photograph.

4. Suite

http://localhost:8123/test/ (tools/devserver.py 8123), browser pane fronted, viewport emulated at 375x812 for runs 2-4.

run	tree	result
1	v26, no emulation	516 / 516
2	v26, 375x812	516 / 516
3	mutation: css/base.css from e5f55fb (v25) under the v26 tests	515 / 516 - exactly "at 360x728 the whole lie grid is above the footer when the burst ends": <i>"the lowest lie button ends at 326 px, the body at 298 px (28 px below the fold)"</i>
4	v26 restored	516 / 516

516, not 511: the fold and movement tests are now run at three page heights instead of two screen heights. No test is RED on purpose. "The deliberate gesture unlocks" passed on every run.

5. The numbers, before and after (suite harness, lock strip reserved)

Measured with the previous mark's "marked - UNDO" banner up, which is the case that moved the grid 78 px before v25. n = 1 run per size per suite run; the values below repeated to the pixel across runs 1-4.

page	v25 grid	v25 body bottom	v25 spare	v26 grid	v26 body bottom	v26 spare
375 x 791	187.2 - 327.2	360.7	+33.5	177.2 - 317.2	405.2	+88.0
360 x 759	187.2 - 327.2	328.7	+1.5	177.2 - 317.2	373.2	+56.0
360 x 728 (his phone)	187.2 - 327.2	297.7	-29.5	177.2 - 317.2	342.2	+25.0

The grid moved **0.0 px** between the running card and the saved card at every size, under both builds. Every lie label fits its button (0 px spill) and the rightmost button ends at 257 px with the lock strip starting at 274.

The two heights the job named - 759 and 791 - are a 21 px status bar and no gesture bar. 728 is what his phone actually gives the app, so that is the size this is built to; 759 passed under v25 by 1.5 px, which is why it could not be the bar.

6. What was clicked, at 360 x 728 in the real app

?sim=1, one tab, Veenker back nine, **hole 10, par 5, 473 yd** - the hole in his photograph. Auto-lock pushed out, LOCK tab up (has-lock-tab on, tab at x = 284). Tee shot marked, then shots 2, 3, 4 and 5 marked and lied (FAIRWAY, ROUGH, SAND), each new mark pressed while the previous mark's banner was still up. n = 3 measured bursts.

	reading
footer	385.6 px (430.3 under v25)
body	131.3 - 342.4
lie grid	176.2 - 316.2, running card and saved card identical (moved 0.0 px)
spare below the grid	26.3 px
rightmost lie button	253.3, lock strip starts at 274
label spill	0 px
hint	"At your ball: MARK SHOT 6.", one line, 18.9 px

The tee shot's burst has no lie grid at all - the tee lie is known, nothing is asked - so the two-line tee hint costs nothing.

The hint is the one thing in that footer whose height is not fixed

At 360 px with the lock strip reserved, measured in the app:

hint	lines	body bottom	spare under the grid
"At your ball: MARK SHOT 3."	1	342.4	+26.3
"Hole 10 is done. Move to hole 11."	1	342.4	+26.3 (no card in this state)
"On the tee: MARK TEE SHOT before you hit."	2	323.6	+7.4 (no lie grid in this state)
"Cup marked. Putt out, then enter the putts and how long the first one was."	2	323.6	+7.4
"On the green: MARK CUP when you walk behind the hole, then enter the putts."	3	304.7	-11.5

7. What changed

file	change
css/base.css	new .screen.play block: footer gap 14 -> 12, footer padding-top 8 -> 6, hint margin-bottom 4 -> 0, footer buttons 60 -> 52 / primary 76 -> 64 / huge 88 -> 80; body padding-top 12 -> 8. Capture card padding 10 -> 8, lie field padding 6/8 -> 4/6, lie label margin 6 -> 4
js/ui/screen-play.js	the screen root is screen play, so the above is the play screen only
test/run.js	the harness restates the footer's 78dvh clamp from its own page height (Section 3); the group runs at 375x791, 360x759 and 360x728 instead of 375x812 and 360x780
js/data/build.js	BUILD.id v25 -> v26 , date 2026-09-16
sw.js	gt-shell-v25 -> gt-shell-v26

REVISION is untouched - that is his call when a build is about to be played. Nothing in js/gps/, js/util/geo.js, js/data/schema.js or js/analysis/ was touched, and the pocket lock, the marks, the hole windows and round save are as v25 left them.

8. His call

1. **The push.** Not taken. v26 is committed and not pushed.
2. **The two long hints.** "On the green: MARK CUP when you walk behind the hole, then enter the putts." wraps to three lines at 360 px and costs the bottom row 11.5 px in the one state that can show it with a card up; "Cup marked. Putt out, then enter the putts and how long the first one was." wraps to two and leaves 7.4 px. Shorter sentences fix it outright; they are his words, so nobody else shortens them.
3. **One photograph after the push,** the same one as before: MARK TEE SHOT, wait for "Tee shot marked", MARK SHOT 2. All six lies should be clear of the footer with about a finger's width of the card's "Captured" line showing under them. If any of it is cut, the number to send back is what the page height is, and the build stays off the course.

9. v27: the hints

Matt, 2026-09-16, on the residual in Section 8 item 2: **"shorten it and push."** Built at 193eb00, tree clean at the spawn.

The two strings

	before	after
on the green	On the green: MARK CUP when you walk behind the hole, then enter the putts.	On the green: MARK CUP, then putts.
cup marked	Cup marked. Putt out, then enter the putts and how long the first one was.	Cup marked. Putt out, then the putts and how long the first was.

Nothing else moved. No CSS was touched — the footer, the card and the lie grid are as v26 left them; only the sentence above them is shorter.

What it is worth, measured in the app at 360 x 728

?sim=1, one tab, viewport emulated 360 x 728, hole 10 par 5 473 yd — the same hole and the same page height as Section 6. Footer **385.6 px**, body 131.3 - 342.4, lie grid bottom **316.2**, all identical to Section 6, so the only variable is

the hint. The hint is 246 px wide with the lock strip reserved; each line is 18.9 px. $n = 1$ run per string, the four strings swapped into the same live footer back to back.

hint	lines	spare under the lie grid
old, on the green	3	-11.5 px — the bottom row under the footer
new, on the green	1	+26.3 px
old, cup marked	2	+7.4 px
new, cup marked	2	+7.4 px

So the state that was 11.5 px short now clears by **26.3 px** — the same number the "At your ball: MARK SHOT 6." state reads in Section 6, which is the best this layout gives. The rightmost lie button still ends at **253.3 px** against a strip starting at 274, and label spill is **0**.

The one-line budget is 246 px, about 35 characters. Fable's suggested On the green: MARK CUP, then the putts. (39) measures two lines; dropping "the" is what buys the line. MARK CUP behind the hole, then putts. (37) is also two lines — the capitals are wide — so "walk behind the hole" could not be kept at any length and is what went.

Why the "cup marked" hint stayed at two lines

One line is reachable (Cup marked. Putt out, then the putts. measures 18.9 px and +26.3) but only by dropping *how long the first one was*, and that is the one thing on that screen he has to notice **before** he picks the ball out — the putts entry asks for the first putt's length after the fact. The job allowed two lines for this hint because there is no state that shows it with a lie grid under it; at two lines it costs the same 37.8 px the old one did, so nothing was given up to keep it. It is shortened from 74 to 64 characters, in his words.

The test

test/run.js, the fold group. The helper now plays one more move at each of the three page heights: shot 2's lie answered **GREEN**, then **MARK SHOT 3** pressed instead of MARK CUP — the one state that puts the "on the green" hint on screen with a capture card under it — and a new assertion per size holds the whole lie grid above the body's bottom edge under that hint.

Proven against the defect first. With js/ui/screen-play.js mutated back to the two old strings and nothing else changed, the suite reads **518 / 519**, failing exactly one test:

at 360x728 the longest hint still leaves the whole lie grid above the footer — under "On the green: MARK CUP when you walk behind the hole, then enter the putts." the lowest lie button ends at 317 px, the body at 304 px (13 px below the fold)

375x791 and 360x759 stay green under the old strings (the hint is 2 lines at 375 px and the 759 page has 18.2 px to give), which is why 728 is the size the group is held to.

Suite

http://localhost:8123/test/ (tools/devserver.py 8123), browser pane fronted.

run	tree	viewport	result
1	v26 at 193eb00, before the change	360x728	516 / 516
2	v27	360x728	519 / 519
3	mutation: the two old strings back in screen-play.js	360x728	518 / 519 — the new test at 360x728, and only it
4	v27 restored	360x728	519 / 519
5	v27	375x812	519 / 519

519, not 516: the new assertion runs at each of the group's three page heights. No test is RED on purpose.

What changed

file	change
js/ui/screen-play.js	nextStepHint(): the two hint strings above, and a comment saying why the "on the green" one has to stay one line

file	change
test/run.js	the fold group plays GREEN then MARK SHOT and asserts the lie grid is above the fold under the longest hint, at all three page heights
js/data/build.js	BUILD.id v26 -> v27 , date 2026-09-16
sw.js	gt-shell-v26 -> gt-shell-v27

REVISION is untouched. Nothing in js/gps/, js/util/geo.js, js/data/schema.js or js/analysis/ was touched, and no CSS was touched.

His call

1. **The push.** Not taken — v27 is committed and not pushed.
2. **The one photograph** from Section 8 item 3 still stands, and it is now worth one more tap: after MARK SHOT 2, answer the lie **GREEN** and press MARK SHOT 3. The hint should read *"On the green: MARK CUP, then putts."* on one line with all six lies clear of the footer.